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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	3	4	2	3	1	1	1	1	2	3	4	1	2	4	3	3	3	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	2	3	2	2	3	1	3	2	2	1	2	3	1	4	2	4	2
Q.	41	42	43	44	45															
A.	4	4	2	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	1	1	3	2	2	2	3	1	3	1	1	1	4	1	3	1	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	3	1	4	2	4	2	3	3	3	1	3	3	1	3	4	3	4	2
Q.	86	87	88	89	90															
A.	2	2	3	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	1	4	2	2	4	3	4	4	3	2	4	3	4	3	4	2	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	2	2	1	4	1	4	4	4	2	3	2	1	4	1	4	4	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	1	3	1	4	4	3	2	1	2	3	4	1	2	1	1	1	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	1	3	4	4	3	4	1	2	1	1	2	4	2	3	1	3	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	2	4	1	3	2	1	2	4										

SOLUTIONS

PHYSICS

1)

In first case : $\frac{A}{3} = \frac{B}{4} \Rightarrow 4A = 3B$ ...(i)

In second case $\Rightarrow \frac{A+30}{6} = \frac{B+30}{7}$

$\Rightarrow 7A - 6B = -30$ ...(ii)

Solving both equation (i) & (ii)

$A = 30\Omega, B = 40\Omega$

2) For $V_B = 8V$,

Voltage across zener $= \frac{8 \times 2}{3} = 5.33V$

So, zener breakdown will not occur. Hence $I_Z = 0$

For $V_B = 15V \Rightarrow$ Zener breakdown occurs.

Hence, $I_Z = \frac{(15-6)}{1k\Omega} - \frac{6}{2k\Omega} = 9mA - 3mA$

$= 6mA$

3) $I_{D_1} = \frac{15-0.3}{1000} = 14.7mA$, A→Q

$V_{L_1} = I_{D_1} \times R_L = 14.7\ mA \times 900\Omega = 13.2\ V$, B→S

$V_{L_2} = I_{D_2} \times R_L = \left(\frac{15-0.7}{1000}\right) \times 900 = 12.9V$, C→P

$V_R = I_{D_1} \times R = 14.7\ mA \times 100\Omega = 1.47V$, D→R

4)

$i = k\theta \Rightarrow \frac{E}{G+R} = k\theta$

$\Rightarrow \frac{1}{\theta} = \frac{(G+R)k}{2.5} \Rightarrow \frac{1}{\theta} = \left(\frac{k}{2.5}\right)R + \frac{kG}{(2.5)}$ ...(i)

Compare (i) with y = mx + C

Slope $= \frac{k}{2.5} = \frac{1}{4} \Rightarrow k = \frac{2.5}{4} = 6.25 \times 10^{-1} A/div$

5) u = –20cm, v = 20cm;

Least division on graph = 1cm

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{20} + \frac{1}{20} \Rightarrow f = 10$ cm

$\frac{\Delta f}{f^2} = \pm\left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}\right)$

$\Rightarrow \Delta f = \pm\left(\frac{1cm}{400} + \frac{1cm}{400}\right) \times 100$

$\Rightarrow \Delta f = \pm 0.5$ cm

Measured focal length = (10 ± 0.05) cm

6) $\delta_{min} = 2i - A \Rightarrow A = 2i - \delta_{min} = 2 \times 46° - 37° = 55°$

Also, $\delta = i + e - A \Rightarrow 51° = 40° + x - 55°$

$\Rightarrow x = 66°$

7)

For resistance, Ohms law : V = RI, straight line through origin, A → S

For Si diode, knee-voltage = 0.7V conducts in one direction, B → R

For LED, knee-voltage/cut-in-voltage ≅ 1.8V and it conducts in forward bias with glow, C → Q

Capacitor conducts a.c with current leading voltage by 90°, D → P

8) $\tau_{max} = pE = \left(2 \times 10^{-8}\right) \times \left(3 \times 10^{5}\right) = 6 \times 10^{-3}$Nm

9) $\frac{KQ_1}{R_1} = \frac{KQ_2}{R_2}$ (same potential) $\Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2} = \frac{1}{2}$

10) For max. force, charges must be $\frac{Q}{2}$ and $\frac{Q}{2}$

11) $V = \frac{KQ}{R} = 200V \Rightarrow KQ = 200\ R$

$V' = \frac{KQ}{r}$

$V' = \frac{KQ}{2R}$ (for r = 2R)

$V' = \frac{200R}{2R} = 100V$

12) $H = \frac{V^2 t}{R}$

13) $E - i\,r_1 = 0$

$\Rightarrow E - \frac{2E}{r_1 + r_2 + R} r_1 = 0$

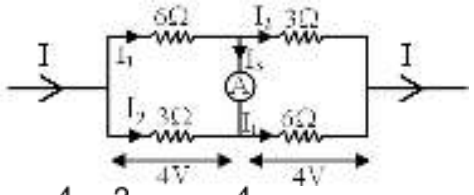
$\Rightarrow R = r_1 - r_2$

14) Maximum power transfer theorem, R = r ⇒ R = 6 Ω

15) $I_{net} = \frac{8}{4} = 2A \Rightarrow I_{6\Omega} = \frac{2}{3}A$ and $I_{3\Omega} = \frac{4}{3}A$

$\Rightarrow I_{wire} = \frac{2}{3}A$

OR



$I_1 = \frac{4}{6} = \frac{2}{3}A$; $I_2 = \frac{4}{3}A$

$I_3 = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}A$

16)

Resistivity of semiconductors decreases with the in crease in temperature.

17) $C = \frac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1} = \frac{4\pi\varepsilon_0 2R\,R}{2R - R}$

$= 8\pi\varepsilon_0 R$

18) By Nodal analysis,

$[(V_A - 10) + (V_A - 5 - 5) + (V_A - 0)] = 0$

$3V_A = 20 \Rightarrow V_A = \frac{20}{3}$

19) $B = \frac{\mu_0 ni}{2R} = \frac{4\pi \times 10^{-7} \times 25 \times 4}{2 \times 5 \times 10^{-2}}$

$= 1.257 \times 10^{-3}$ T

20) $\vec{F} = q(\vec{v} \times \vec{B})$

$F = qvB\sin\theta \Rightarrow B = \frac{F}{qv\sin\theta}$

For minimum magnetic field, θ = 90°

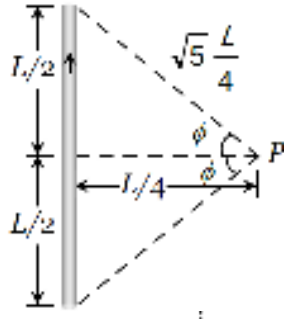
$B_{min} = \frac{F}{qv}$

$\Rightarrow B_{min} = \frac{F}{qv} = \frac{10^{-10}}{10^{-12} \times 10^{5}} = 10^{-3}$ Tesla

(in negative $\hat{z}$-direction)

21) By using

$$B = \frac{\mu_0}{4\pi} \cdot \frac{i}{a}(\sin\phi_1 + \sin\phi_2)$$



$$\Rightarrow B = \frac{\mu_0}{4\pi} \cdot \frac{i}{(L/4)}(2\sin\phi)$$

Also $\sin\phi = \frac{L/2}{\sqrt{5}L/4} = \frac{2}{\sqrt{5}}$

$$\Rightarrow B = \frac{4\mu_0 i}{\sqrt{5}\pi L}$$

22) Pole strength doesn't depend upon the length.

23) Concept of magnetic screening/shielding

24) For r ≤ R : $E(2\pi r) = \pi r^2 \frac{dB}{dt} \Rightarrow E \propto r$;

For r > R : $E(2\pi r) = \pi R^2 \frac{dB}{dt} \Rightarrow E \propto \frac{1}{r}$

25) $\Delta\phi = BA(\cos\theta_2 - \cos\theta_1)$

Here, $\theta_1 = 0°$ and $\theta_2 = 180°$

$\therefore \Delta\phi = BA(\cos\theta_2 - \cos\theta_1)$

 Induced emf = $-\frac{\Delta\phi}{\Delta t} = \frac{2BA}{\Delta t}$

$= \frac{2 \times 0.1 \times 1}{0.01} = 20V$

The coil is closed and has a resistance of 2.0 Ω

Therefore, i = 20/2 = 10A

26)

R_2 is in series with L and battery is connected across series combination of L and R_2.

So, $\tau = \frac{L}{R_2}$

27)

Power $P = \frac{1}{2}V_0 I_0 \cos\phi$

$= 0.5 \times 150 \times 150 \times \cos60°$

$= \frac{22500}{4} = 5625W$

28) $Q = \frac{\omega_r L}{R} = \sqrt{\frac{L}{C}}.\frac{1}{R}$

$Q = \sqrt{\frac{1}{9}}.\frac{1}{3} = \frac{1}{9}$

29) bandwidth (BW) $= \frac{f_r}{Q} = \frac{f_r}{\left(\frac{X_L}{R}\right)}$

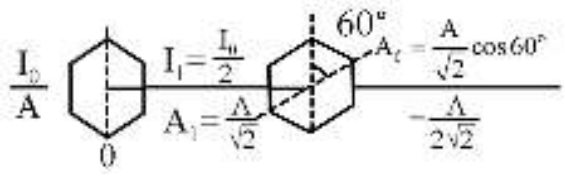
$= \frac{12 \times 10^3 \times 5}{300} = 200\ Hz$

$f_1 = f_r - \frac{BW}{2} = 12000 - 100 = 11900\ \Omega$

$f_2 = f_r + \frac{BW}{2} = 12000 + 100 = 12100\ \Omega$

30) $\beta = \frac{2\lambda D}{a} = \frac{2 \times 500 \times 10^{-9} \times 80 \times 10^{-2}}{0.20 \times 10^{-3}} m$

$= 4 \times 10^{-3} m = 4\ mm$

31) 

32) $B_0 = \frac{E_0}{c} = \frac{60}{3 \times 10^8} = 2 \times 10^{-7} T$

$\hat{E} \times \hat{B}$ must be direction of propagation.

So, $\hat{B} \rightarrow z$ - axis

$k = \frac{2\pi}{\lambda} = \frac{\pi}{4} \times 10^3 m^{-1}$

$E_y = 60 \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t)\right] \hat{j}\ Vm^{-1}$

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t)\right] \hat{k}\ T$

33) $I = \frac{P}{4\pi r^2} = \frac{B_0^2}{2\mu_0} \times C$

$B_0 = \sqrt{\frac{2P\mu_0}{4\pi r^2 C}} = \sqrt{\frac{2 \times 15 \times 10^{-7}}{(2)^2 \times \left(3 \times 10^8\right)}}$

$= 5 \times 10^{-8}\ T$

34) $\frac{\delta_a}{\delta_w} = \frac{(a\mu_g - 1)}{(w\mu_g - 1)} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{3/2}{4/3} - 1\right)} = 4 \Rightarrow \delta_w = \frac{\delta_a}{4}$

35) R = 2f
= 2 × 30 = 60 cm

36) $\sin\theta_C = \frac{1}{\mu}$

$\sin\theta_C = \frac{V}{C} = \frac{\frac{C}{2}}{C}$

$\sin\theta_C = \frac{1}{2}$

$\theta_C = 30°$

37)

Free electrons are at different energy levels in metal, so KE of photoelectrons emitted are not equal even if monochromatic light is incident on metal surface.
Work function (ϕ_0) = $h\nu_0$,
where, ν_0 → Threshold frequency time interval between incident of photon and emission of electron is 10^{-9} s or less in photelectric effect.

38) $K_{max} = \left[\frac{12400}{5000} - 1.2\right] eV$

K_{max} = 1.28 eV and K_{min} = 0
on reaching anode
K'_{max} = (1.28 + 4) = 5.28 eV
K'_{min} = (0 + 4) = 4 eV

39)

A single experiment with light exhibits only one nature of light, either particle nature or wave nature.

40) $v_P > v_A$

$\lambda \propto \frac{1}{v}$

So, $\lambda_P < \lambda_A$

41) $\frac{mv^2}{r} = \frac{KZe^2}{r^2}$ .....(i)

$mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$ .....(ii)

in CGS, K = 1
using (i) & (ii), we get

$$m \times \frac{n^2h^2}{4\pi^2m^2r^2} = \frac{Ze^2}{r}$$

$$\Rightarrow r = \frac{n^2h^2}{4\pi^2mZe^2}$$

42)

The proposed path of e^- in rutherford model of an atom is circular.

43)

For half wave rectifier $V_{o/p} \Rightarrow V_{rms} = \frac{V_0}{2}$

$V_{rms} = \frac{200}{2} = 100$ V

44)

LED is used in forward bias, so, characteristic of LED is same as p-n junction diode.

45)

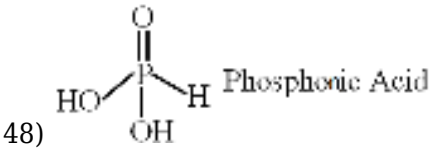
A	B	C
0	0	0
1	1	1
0	1	0
1	0	0
0	0	0
1	1	1
0	0	0

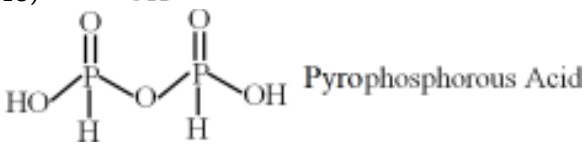
This truth table shows the output (C) for given inputs A and B.
It is an AND gate.

CHEMISTRY

46) Ziese's salt → $K[PtCl_3(C_2H_4)]$
Sodium nitroprusside → $Na_2[Fe(CN)_5NO]$
Prussians Blue → $Fe_4[Fe(CN)_6]_3$
Brown ring test complex → $[Fe(H_2O)_5NO]SO_4$

47) $[Ni(PPh_3)_4]$, $[Co(en)_3Cl_3]$, $[Fe(CO)_5]$ Diamagnetic species

48) 



49) I – White
II – Blue
III – Red
IV – Yellow

50) $PbO_2 > SnO_2 > GeO_2$ Oxidising power order

51) Order of stability / E_{absorb}

$[Co(EDTA]^- > [Co\ (dien)_2]^{+3} > [Co(en)_3]^{+3}$

No. of chelating 5 4 3

52)

Ar is major constituent in atmosphere.

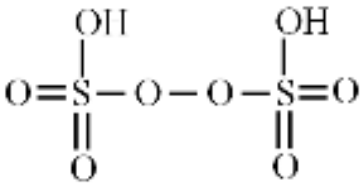
53) $K_2SO_4.Al_2(SO_4)_3.24H_2O$ Potash alum
colourless

54)

$[Cr(gly)_3]$ → show G.I. & O.I.
$[Fe(gly)(NH_3)_2Cl_2]$ → show G.I. & O.I.

55) $PbCl_2$ (white) soluble in HOT water.

56) NO (Nitric oxide) = Neutral
N_2O_3 (Nitrogen sesquioxide) = Acidic

57) 

58) $Na_2\left[\overset{+2}{Fe}(CN)_5NO\right] + Na_2S \rightarrow Na_4\left[\overset{+2}{Fe}(CN)_5NOS\right]$

$\overset{+2}{CuSO_4} + KI \longrightarrow \overset{+1}{Cu_2I_2} + I_2 + K_2SO_4$

$\overset{+3}{Fe}Cl_3 + KI \rightarrow \overset{+2}{Fe}I_2 + I_2 + KCl$

$\overset{+2}{Fe}SO_4(aq) + NO \rightarrow \left[\overset{+1}{Fe}(H_2O)_5NO\right]SO_4$

59)

$Al(OH)_3$ adsorb blue colour from the solution & forms insoluble adsorption complex named 'Lake'.

60)

(1) K_2MnO_4 → Only show d-d transition.
(2) In basic medium V_2O_4 give VO_3^-.

61)

K1 = K2

$10^{16}\, e^{-\frac{3000}{T}} = 10^{15} e^{-\frac{2000}{T}}$

$10 = e^{\frac{1000}{T}}$

$\ell n10 = \frac{1000}{T}$

$T = \frac{1000}{\ell n10} = 434.22$ K

62)

For ideal solution $\Delta S_{mix} > 0$

63)

$\pi = iCRT$

$\pi = 5 \times 0.2 \times 0.0821 \times 300$

$= 24.6$ atm

64)

$\alpha = \frac{\Lambda_M}{\Lambda_M^\infty} = \frac{38.8}{388} = 0.1$

$\Lambda_M^\infty = 425 + 91 - 128 = 388$

65)

$r = K[A]\,[B_2]$

$K_1[A_2] = K_2[A]^2$

$[A] = \sqrt{\frac{K_1}{K_2}[A_2]}$

$r = K\sqrt{\frac{K_1}{K_2}[A_2]} \cdot [B_2]$

$r = K'[A_2]^{1/2}[B_2]$

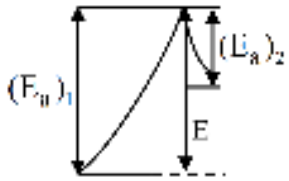
Order = 1.5

66)

$t = \frac{2.303}{4.62 \times 10^{-3}} \log \frac{0.4}{0.1}$

t = 300 sec.

67)



$(E_a)_1 > (E_a)_2$

E_T same for both

68)

From data r = K[A] [B]

$K = \frac{2 \times 10^{-6}}{10^{-2} \times 2 \times 10^{-2}} = 10^{-2}$ L/mol sec

69)

$a = \frac{\Lambda_M}{\Lambda_M^\infty}$

$\Lambda_M = 0.1 \times 390 = 39 = \frac{K \times 1000}{M}$

$K = \frac{39 \times .001}{1000} = 3.9 \times 10^{-5}$

70)

$\Box n\ K = \Box nA - \frac{E_a}{RT}$

$-\frac{E_a}{R} = -3 \times 10^{-5}$

$E_a = 3 \times 8.3 \times 10^{-5}$
$= 2.5 \times 10^{-4}$

71)

$5 = \frac{45}{180} \times RT$
$T = \frac{20}{R}$
$2 = C \times RT$
$2 = C \times R \times \frac{20}{R}$
$C = \frac{2}{20} = 0.1$ mol/L

72)

$.05 = \frac{5}{M_B} \times \frac{1.5}{300} \times 1000$
$M_B = 500$

73)

$H_{2A} \rightarrow 2H^+_A + 2e^- \quad E^0_{cell} = 0$
$2H^+_C + 2e^- \rightarrow H_{2C}$
$E_{cell} = 0 - \frac{RT}{2F} \log_e \frac{P_{H_2C}}{P_{H_2A}}$
$= -\frac{RT}{2F} \log_e \frac{P_2}{P_1}$
$= \frac{RT}{2F} \ell n \frac{P_1}{P_2}$

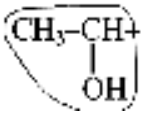
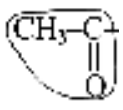
74)

Assertion is true & Reason is false
In galvanic cell electron flow from anode to cathode in external circuit.

75)

FACT

76)

Compounds having  and  groups form haloform when heated with $X_2/\overset{\ominus}{O}H$.

77)

[Rate of $SN^1 \propto$ stability of $C^{\oplus}$]

78) NCERT Pg # 358

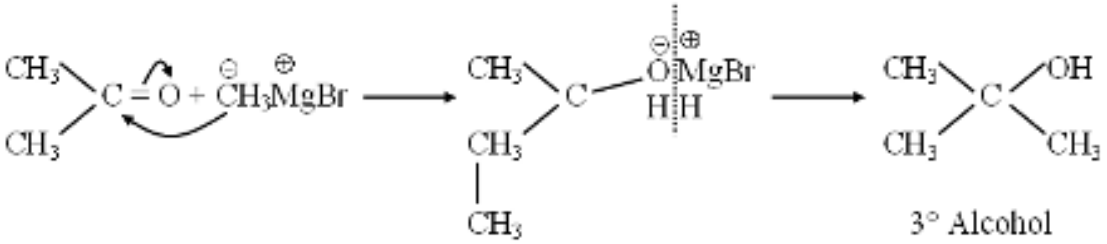
Explain question : Identify which of the given carbonyl compounds, when reacted with CH_3MgBr, will produce a tertiary alcohol?

Concept : This question is based on Nucleophilic addition reaction.

Addition of RMgX.

Solution :

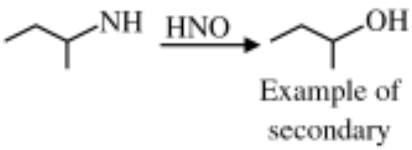
When ketone reacted with Grignard reagent, it will product tertiary alcohol.



So ketone (acetone) gives tertiary alcohol on react with CH_3MgBr.

Final answer : option 3

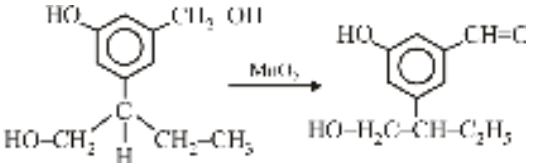
79)



80) Rate towards lucas reagent $\propto$ [Stability of $C^{\oplus}$]

$3°C^{\oplus} > 2°C^{\oplus} > 1°C^{\oplus}$

81)

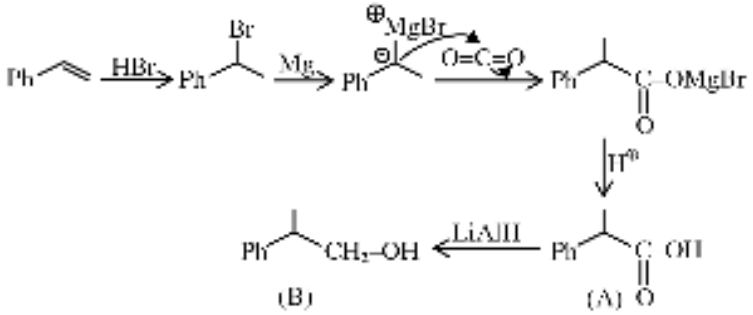


only benzyl alcohol can be oxidised.

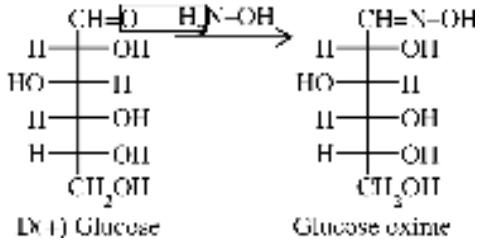
82)



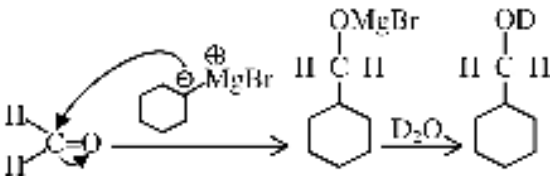
83)



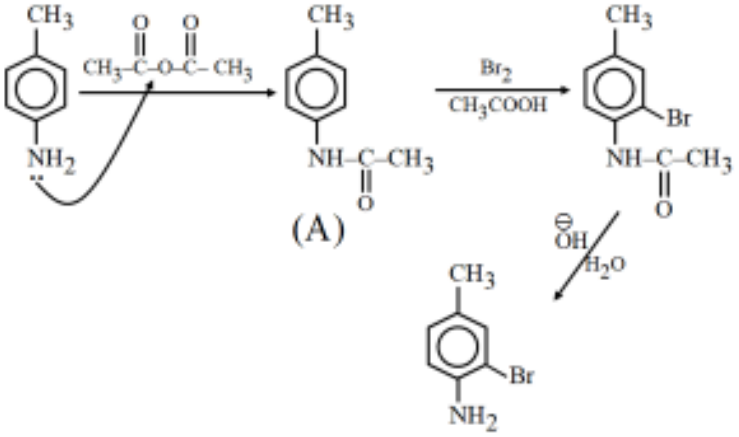
84)



85)

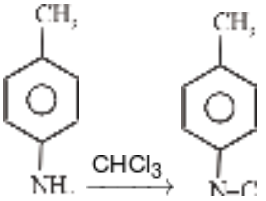


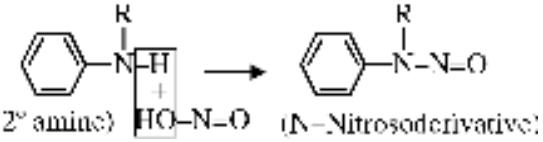
86) **Explain the question :** Identify the final product 'C' formed in given reaction sequence?
Concept : This question is based on acylation, electrophilic substitution reaction, bromination, hydrolysis.
Solution :



Final answer : 2

87)



88) 

(2° amine) (N–Nitrosoderivative)

89)

(A) → R–NH_2
(B) → R–CH_2–NH_2
Both are homologues

90)

A. Aniline is used in making azo dyes by coupling reaction.
B. Nitrobenzene is used as solvent in Friedal crafts reaction bar nitrobenzene not show friedal craft reaction.
C. Sulphanilamide is parent compound for the formation of sulpha drugs.
D. Trinitrotoluene (TNT) used as explosive.

BIOLOGY

91) NCERT-XII Pg # 69

92) NCERT-XII, Pg. # 92

93)

NCERT-XII, Page # 62,69

94) NCERT-XII Pg. # 81

95) NCERT-XII, Pg. # 59

96) NCERT XII Pg.# 67,68

97) NCERT XII Pg. # 83 (Linkage and Recombination)

98) NCERT-XII, Pg. # 73, 74, 75, 76

99)

NCERT-XII, Pg. # 92

100) NCERT Pg. # 90 (Mendelian Disorder)

101) NCERT XII, Pg # 97, 98

102) NCERT XII, Pg # 90, 91

103) NCERT XII, Pg # 90, 91

104) NCERT XII, Pg # 83, 84

105)

NCERT-XII, Pg. # 98

106) The size of VNTR varies in size from 0.1 to 20 kb- NCERT 11TH

107) NCERT-XII Pg. # 122

108) NCERT Pg. # 96 (Structure of Polynucleotide Chain)

109) NCERT-XII, Pg. # 112

110)

NCERT-XII, Pg. # 104, 107, 114

111)

NCERT-XII, Pg. # 98

112)

NCERT-XII, Pg. # 230

113)

NCERT-XII, Pg. # 238

114)

NCERT-XII, Pg. # 232

115)

NCERT-XII, Pg. # 233

116)

NCERT-XII, Pg. # 232-238

117)

NCERT-XII, Pg. # 259

118)

NCERT XII Pg # 265

119) NCERT Pg # 267

120)

NCERT-XII, Pg. # 266

121) NCERT-XII Pg # 207

122)

NCERT-XII Pg#207

123)

NCERT-XII, Pg. # 243

124) NCERT, Pg # 208

125)

NCERT-XII, Pg. # 261

126)

NCERT-XII, Pg. # 247

127) NCERT Pg. # 13,18,19,22

128) NCERT XII Pg. # 13

129)

Question Explanation: The question is asking about the correct match

Concept: Seed and fruit

Solution: Scutellum is the single cotyledon in the grass family.

Final answer: Option 4

130) NCERT-XII, Pg. # 13

131) NCERT XII, Pg. # 33-38

132)

Answer : Sporopollenin

133) XII NCERT PAGE NO.24

134)

NCERT Pg.# 18

135) NCERT Pg.# 21, 22

136)

NCERT-XII, Pg. # 198

137)

NCERT-XII, Pg. # 199

138)

NCERT-XII, Pg. # 200

139) NCERT Pg. No. # 200

140) NCERT Pg. # 166

141)

NCERT-XII, Pg. # 195-204

142)

NCERT XII Pg # 170

143) NCERT Pg. # 172-173, 182-183

144)

NCERT-XII, Pg. # 208

145) NCERT Pg. # 180,181,182 & 183

146)

NCERT-XII, Pg. # 214

147) NCERT Pg. No. # 185

148) NCERT Pg. # 179, 180, 182, 183

149)

NCERT-XII, Pg. # 184

150) NCERT Pg # 158

151)

NCERT Pg # 153

152) NCERT, Pg. # 155-156

153)

NCERT XII, Page # 54

154) NEW NCERT-XII, Pg. # 31 (E), 33 (H)

155)

NCERT XII, Pg # 36

156)

NCERT XII, Pg # 32, 33, 34

157) Ampulla of fallopian tube

158) Placenta .

159)

NCERT XII, Pg # 48

160)

NCERT XII, Pg # 37

161)

NCERT XII, Pg # 38

162) 12 week .

163)

NCERT XII, Pg # 45, 46

164)

NCERT XII, Pg # 47

165)

NCERT-XII, Pg. # 44

166) NCERT-XII Pg#139

167) NCERT PG. # 142, 143

168) NCERT Pg. # 134 and 138

169) NCERT Pg # 160

170)

NCERT XII, Pg # 141

171)

NCERT XII, Pg # 136

172)

NCERT XII, Pg # 141

173)

NCERT-XII, Pg. # 142-143

174)

NCERT XII, Pg # 116, 117, 118

175) NCERT-XII, Pg # 122

176) NCERT Pg # 124, 125

177) NCERT Pg. # 119

178)

Question Explanation – The question is asking for the name of the process in which new species originate from a single common ancestor.
Concept – This question is based on Speciation
Solution – When populations derived from a common ancestor undergo evolutionary changes and become reproductively isolated, they form new species. This process of formation of new species from a common ancestor is called speciation. Genetic drift, mutation, and natural selection are mechanisms that contribute to evolution but do not directly define the formation of new species.
NCERT Pg. # 119
Final Answer – Option 1

179)

NCERT-XII, Pg. # 116

180) NCERT Pg # 116